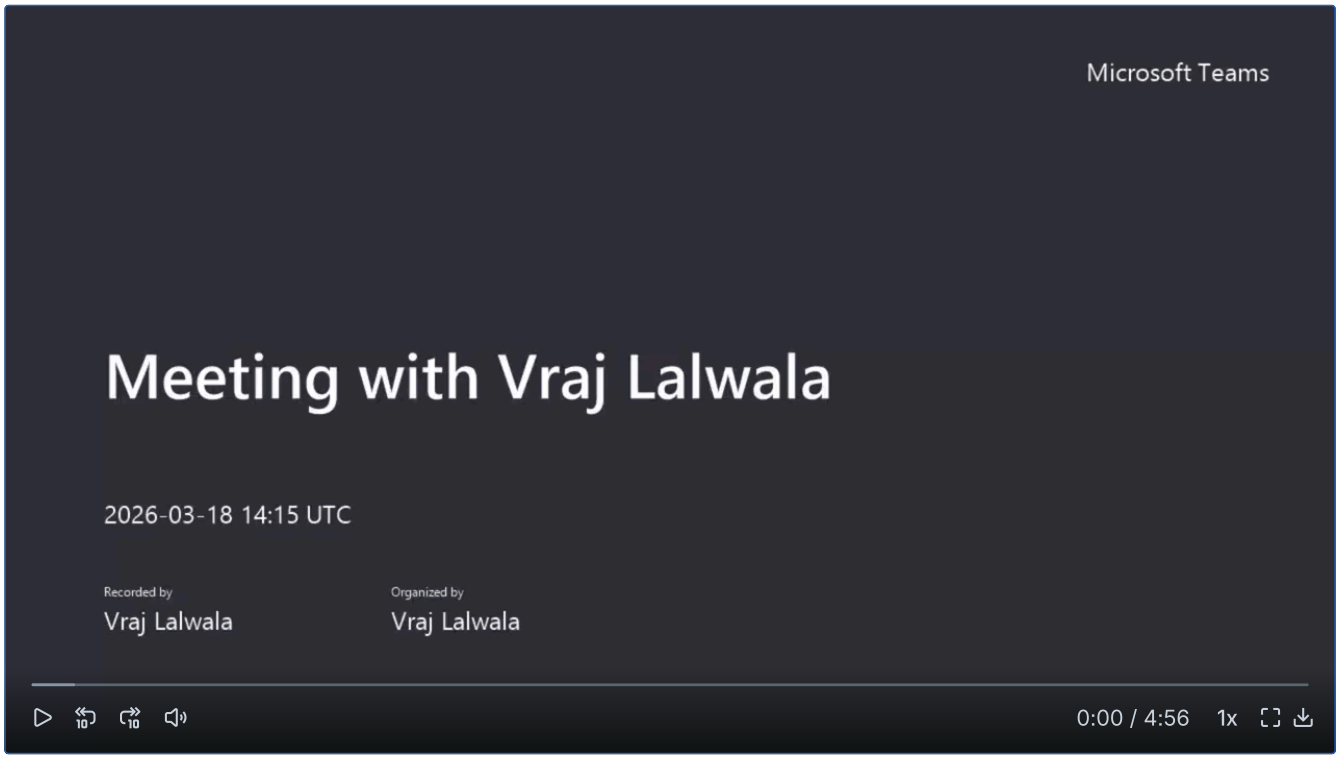
SQL vs NoSQL Databases in Bookstore API Deployment



1. Introduction

In the deployment of the **Bookstore API application**, the database layer plays a critical role in storing and retrieving application data such as book records.

The application is deployed using a **2-tier architecture** where:

- The **application layer** is hosted on **Amazon EC2**
- The **database layer** is hosted using **MongoDB Atlas**

MongoDB is a **NoSQL database**, which differs significantly from traditional SQL databases.

This document explains:

- What SQL and NoSQL databases are
- Their differences
- Why NoSQL was selected for this project

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2. Role of Database in the Bookstore API

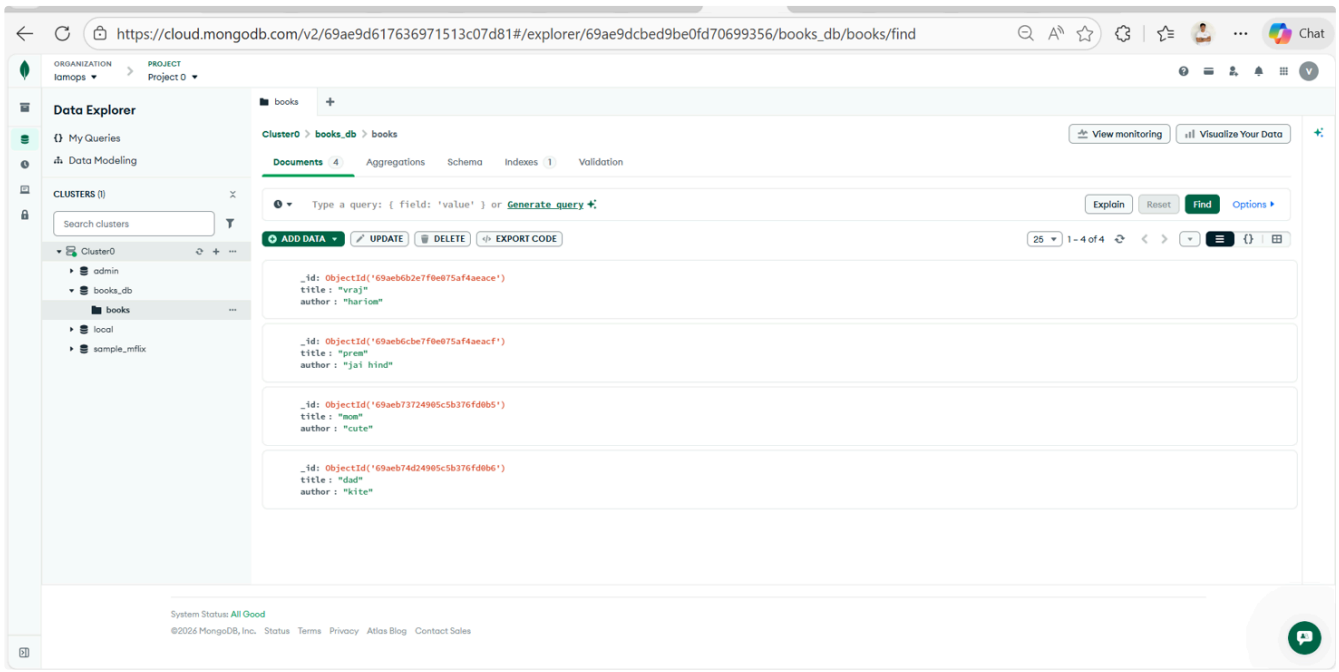
The Bookstore API application performs operations such as:

- Adding book records
- Retrieving book information
- Updating book details
- Deleting book records

These operations require a database that can:

- Store book data efficiently
- Handle API requests
- Support scalable application architecture

In this project, **MongoDB** was used because it supports flexible data storage and integrates easily with modern web applications.



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3. SQL Databases

SQL databases are **relational databases** that store data in structured tables consisting of rows and columns.

Examples of SQL databases include:

- MySQL
- PostgreSQL
- Microsoft SQL Server

In SQL databases, the structure of the data must be defined before inserting records.

Example table for the bookstore application:

BookID	Title	Author	Price
1	DevOps Guide	John	500

Example SQL query:

```
INSERT INTO books (title, author, price)
VALUES ('DevOps Guide', 'John', 500);
```

SQL databases are widely used in systems that require strict data consistency such as:

- Banking systems
- Financial systems
- ERP systems

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4. NoSQL Databases

NoSQL databases are **non-relational databases** designed to store large volumes of flexible data.

Instead of tables, NoSQL databases store data in formats such as:

- JSON documents
- Key-value pairs
- Graph structures
- Column families

The database used in this project is:

- MongoDB

MongoDB stores data in **JSON-like documents**, which makes it highly suitable for API-based applications.

Example MongoDB document used in this project:

```
{
 "title": "DevOps Guide",
 "author": "John",
 "price": 500
}
```

Unlike SQL databases, MongoDB does not require a predefined schema.

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5. Key Differences Between SQL and NoSQL

Feature	SQL Databases	NoSQL Databases
Data Structure	Tables with rows and columns	JSON-like documents
Schema	Fixed schema	Flexible schema
Scalability	Vertical scaling	Horizontal scaling
Query Language	SQL	API-based queries
Data Relationships	Strong relational support	Limited relational support
Performance	Good for structured data	Good for large-scale applications
Examples	MySQL, PostgreSQL	MongoDB, Cassandra

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6. SQL vs NoSQL in the Bookstore API Architecture

In this project, a **2-tier architecture** was implemented.

Architecture:

```
User
   |
   | API Request
   ▼
FastAPI Application
(EC2 Instance)
   |
   | Database Connection
   ▼
MongoDB Atlas Database
```

The FastAPI application sends requests to the MongoDB database to store and retrieve book information.

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7. Why NoSQL Was Selected for This Project

Several factors influenced the choice of MongoDB over a SQL database.

Flexible Data Model

Book records may contain varying attributes such as:

- title

- author
- price
- category
- publisher

MongoDB allows flexible schemas without altering database structure.

Better Integration with APIs

The Bookstore API uses JSON requests and responses.

MongoDB stores data in JSON-like documents, which simplifies integration.

Scalability

NoSQL databases support **horizontal scaling**, allowing the system to handle large traffic volumes in future deployments.

Faster Development

Developers can modify application data structures without changing database schema.

This speeds up development and testing cycles.

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8. Practical Example from the Project

Example API request used in the project:

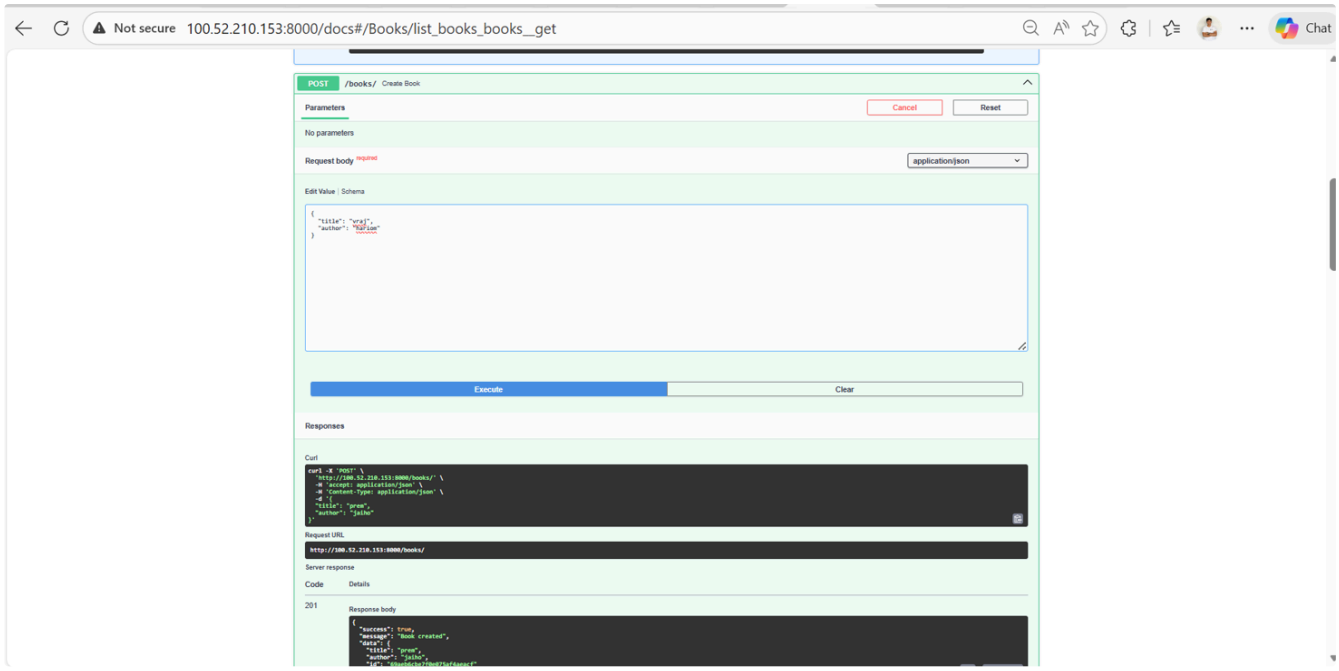
Endpoint:

```
POST /books
```

Request Body:

```
{
 "title": "hariom",
 "author": "vraj"
}
```

This data is stored in MongoDB as a document.



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9. Advantages of SQL and NoSQL

Advantages of SQL

- Strong data consistency
- ACID transaction support
- Mature ecosystem
- Ideal for complex relational queries

Advantages of NoSQL

- Flexible schema
- High scalability
- Handles large data volumes
- Faster development for modern applications

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10. Conclusion

In the Bookstore API deployment project, the database layer was implemented using MongoDB, a NoSQL database.

The use of MongoDB allowed:

- Flexible data storage
- Easy integration with FastAPI

- Scalable architecture

While SQL databases remain essential for systems requiring strict relational data consistency, NoSQL databases provide better flexibility and scalability for modern API-based applications.

Therefore, NoSQL was the preferred choice for this project's **2-tier architecture deployment on AWS**.